

Tech products & brand perception: Public opinion & sentiment analysis project

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Introduction

In the technology sector, consumer perception is critical for brand reputation, yet user experiences are often shared across dispersed and unstructured online platforms. The main challenges for this project include handling noisy data, sarcasm, and incomplete reviews, as well as managing content in multiple languages like English, French, and Indonesian.

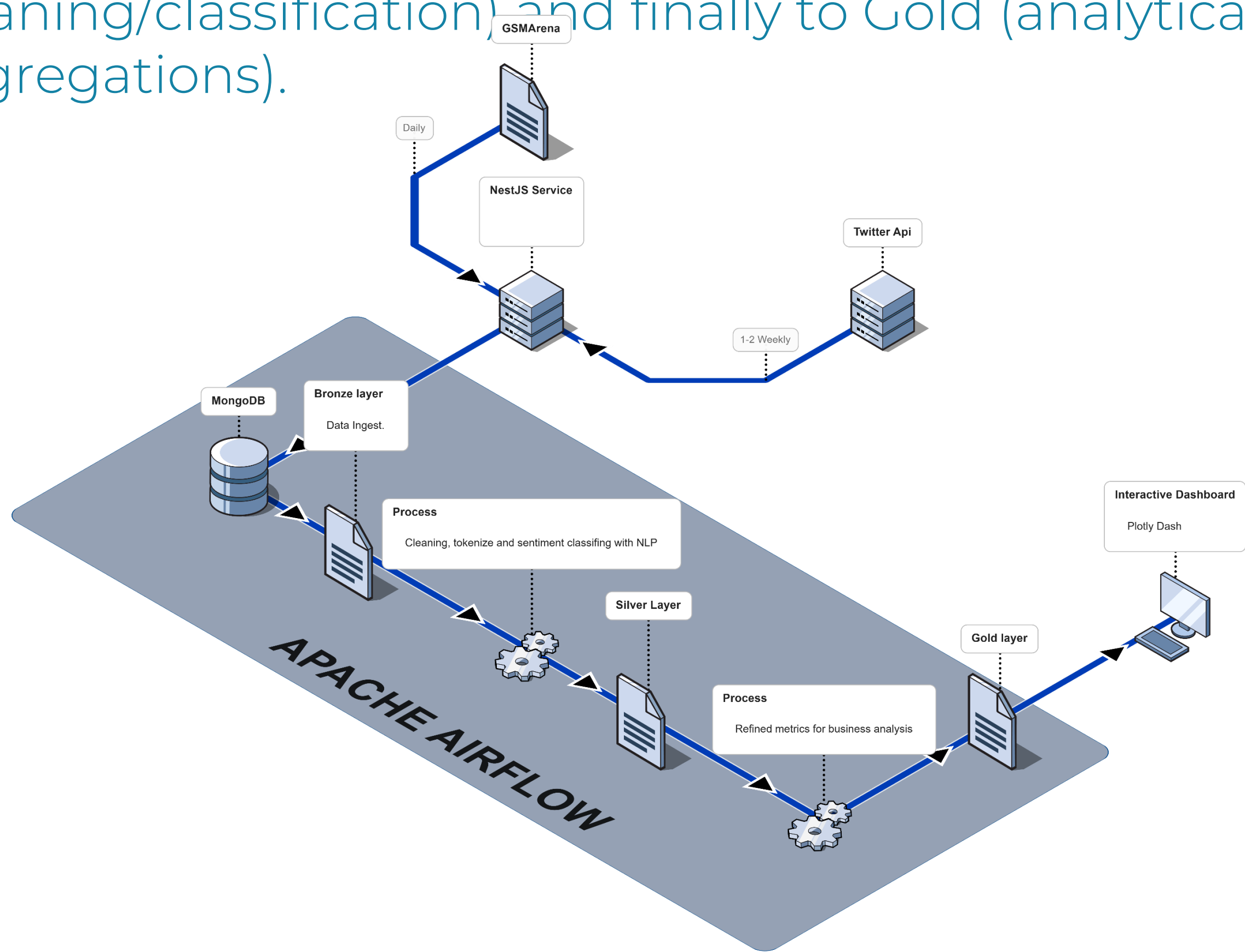
Goal

To design implement an automated end-to-end data pipeline that integrates social media and specialized reviews to provide actionable sentiment insights through an interactive dashboard

Proposed Solution & Architecture

The system follows a Medallion Architecture with a newly implemented automated extraction layer:

- **Automated web scraping:** A dedicated service build with NestJS automates daily data collection. It utilizes Axios for requests, Cheerio for parsing, and Github Actions for cyclic scheduling.
- **Storage strategy:** All scraped data is persistently stored in a MongoDB database for reliability.
- **Ingestion frequencies:** To optimize token expenditure and budget constraints, the API ingestion is scheduled 1-2 times per week, while the cost-effective web scraping runs daily.
- **Data layers:** Data flows from Bronze (raw JSON) to Silver (NLP cleaning/classification) and finally to Gold (analytical aggregations).



Technologies

- **Backend & Automation Web scraping and API requests:** NestJS, Axios, Cheerio, Github Actions.
- **Data Processing:** Python (Pandas) and NPL libraries.
- **Orchestation:** Apache Airflow (DAGs).
- **Database:** MongoDB (Raw Storage)
- **Visualization:** Plotly Dash.
- **Repository:** <https://github.com/Aletarget/Data-Analysis-Programming>



Experiments & Validation

- **Automation Test:** Validation of the NestJS service to ensure cyclic execution without manual intervention.
- **Source Complementarity:** Successfully integrates X/Twitter (broad real-time reactions) with GSMarena (deep technical reviews)
- **Data Quality:** Handled multi-language content (English, French, Indonesian) and informal text through preliminary cleaning.

Results & Discussion

Efficiency: The automated scraping reduces manual workload and ensures a steady data flow despite API tokens limitations.
Multi-dimensional view: The pipeline allows comparing speculative social media “buzz” against stable, user-experience-based reviews.
Adaptability: The project demonstrated high flexibility by pivoting from Reddit to X and adjusting frequencies to manage costs.

Conclusion

The project successfully established a scalable and cost-efficient architecture. By automating the scraping process and optimizing API usage, the system effectively captures public sentiment while remaining within technical and budgetary constraints.